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Laporan Akhir

Praktikum Jaringan Komputer

Vlan Trunk OSPF MultiVendor

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2026

1 Langkah-Langkah Percobaan

Praktikum Modul 5 mengangkat tema VLAN, Trunk, dan OSPF MultiVendor. Pada praktikum ini, dilakukan konfigurasi perangkat Cisco, Huawei, dan MikroTik untuk mengatur segmentasi jaringan lokal serta distribusi informasi *routing* dinamis.

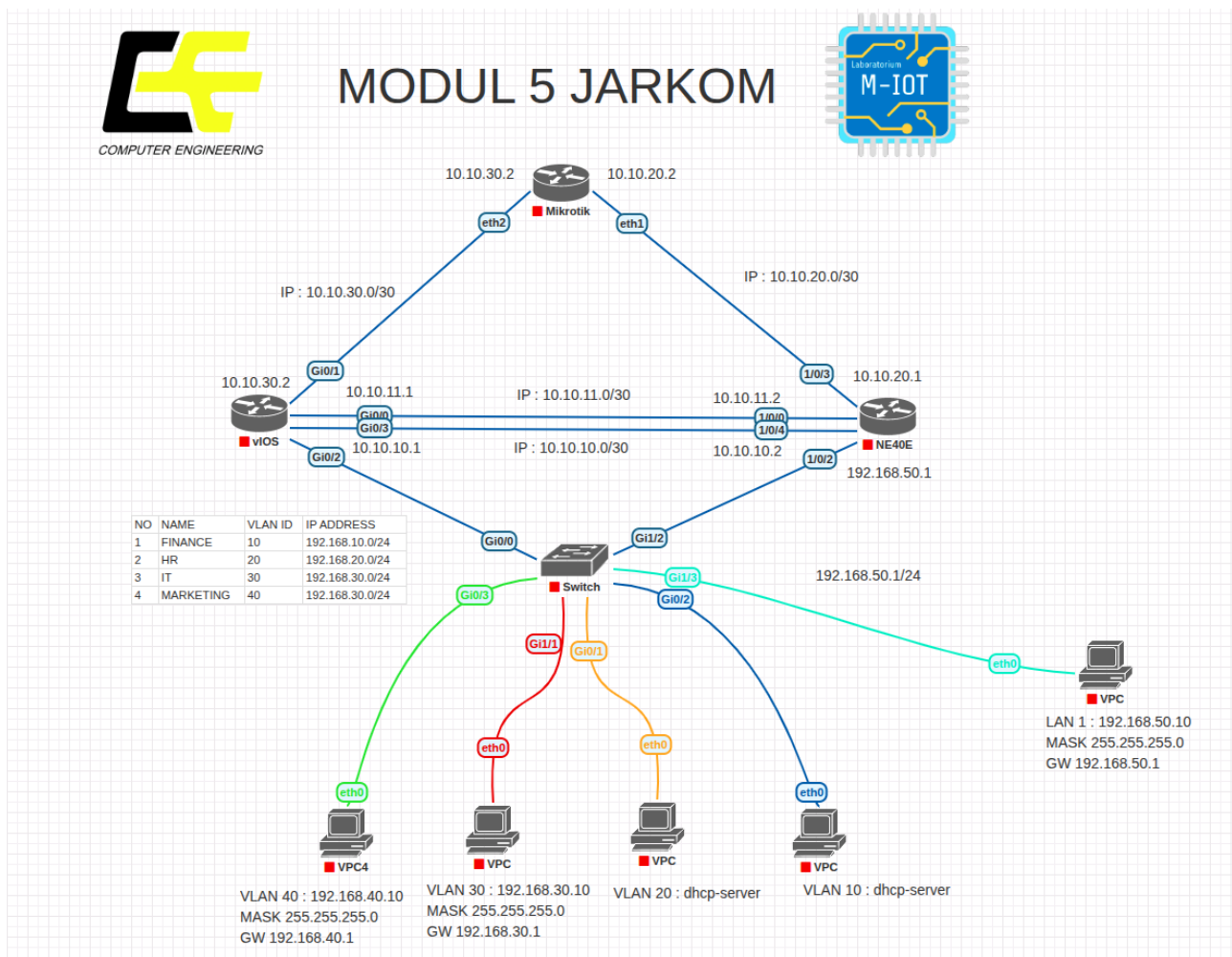
1.1 Topologi Jaringan

1. Topologi Jaringan

Susunan topologi jaringan dibuat pada simulator sesuai dengan panduan modul.

DISCLAIMER: Terdapat penyesuaian pada dua *link* jalur Cisco-Huawei di topologi:

- g0/0 terhubung ke Ethernet 1/0/4
- g0/3 terhubung ke Ethernet 1/0/0



Gambar 1: Topologi Modul 5

2. Tabel Addressing VLAN

VLAN ID	Nama VLAN	Network	Gateway	Keterangan
10	FINANCE	192.168.10.0/24	192.168.10.1	DHCP Server
20	HR	192.168.20.0/24	192.168.20.1	DHCP Server
30	IT	192.168.30.0/24	192.168.30.1	Static IP
40	MARKETING	192.168.40.0/24	192.168.40.1	Static IP

3. Tabel Addressing Client

Perangkat	VLAN	IP Address	Gateway	Keterangan
PC Finance	10	DHCP	192.168.10.1	Mendapat IP otomatis dari DHCP Server
PC HR	20	DHCP	192.168.20.1	Mendapat IP otomatis dari DHCP Server
PC IT	30	192.168.30.10/24	192.168.30.1	IP static
PC Marketing	40	192.168.40.10/24	192.168.40.1	IP static
PC LAN Huawei	-	192.168.50.10/24	192.168.50.1	LAN tambahan pada sisi Huawei

4. Tabel Link Antar Router

Link	Network	Perangkat	Interface	IP Address
Cisco ↔ Huawei Link 1	10.10.10.0/30	Cisco Router	Gi0/3	10.10.10.1/30
Cisco ↔ Huawei Link 1	10.10.10.0/30	Huawei Router	Ethernet1/0/0	10.10.10.2/30
Cisco ↔ Huawei Link 2	10.10.11.0/30	Cisco Router	Gi0/0	10.10.11.1/30
Cisco ↔ Huawei Link 2	10.10.11.0/30	Huawei Router	Ethernet1/0/4	10.10.11.2/30
Cisco ↔ MikroTik	10.10.30.0/30	Cisco Router	Gi0/1	10.10.30.1/30
Cisco ↔ MikroTik	10.10.30.0/30	MikroTik	ether2	10.10.30.2/30
MikroTik ↔ Huawei	10.10.20.0/30	MikroTik	ether1	10.10.20.2/30
MikroTik ↔ Huawei	10.10.20.0/30	Huawei Router	Ethernet1/0/3	10.10.20.1/30

1.2 Praktikum 1: Konfigurasi Cisco Switch

Pada tahap awal, Cisco Switch dikonfigurasi untuk membuat VLAN, menentukan port akses ke client, dan membuat jalur trunk menuju router. Ketika masuk pertama kali dan muncul dialog inisialisasi, opsi no dipilih.

1. Masuk ke Terminal Cisco Switch

Mengaktifkan mode *privileged EXEC* dan masuk ke mode konfigurasi global.

```
Switch> enable
Switch# configure terminal
Switch(config)#
```

2. Pembuatan VLAN 10, 20, 30, dan 40

Mendaftarkan ID dan nama masing-masing segmen VLAN pada switch.

```
vlan 10
  name Finance
exit

vlan 20
```

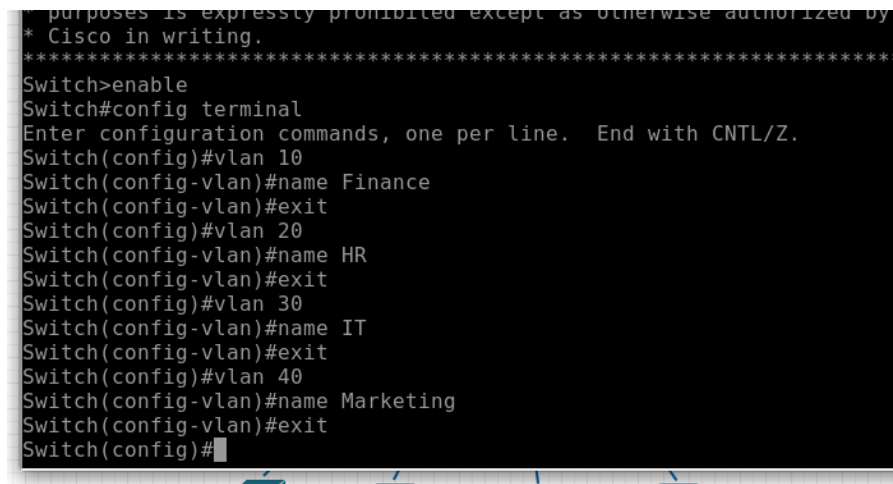
```

name HR
exit

vlan 30
name IT
exit

vlan 40
name Marketingg
exit

```



```

Switch>enable
Switch#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name Finance
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name HR
Switch(config-vlan)#exit
Switch(config)#vlan 30
Switch(config-vlan)#name IT
Switch(config-vlan)#exit
Switch(config)#vlan 40
Switch(config-vlan)#name Marketingg
Switch(config-vlan)#exit
Switch(config)#

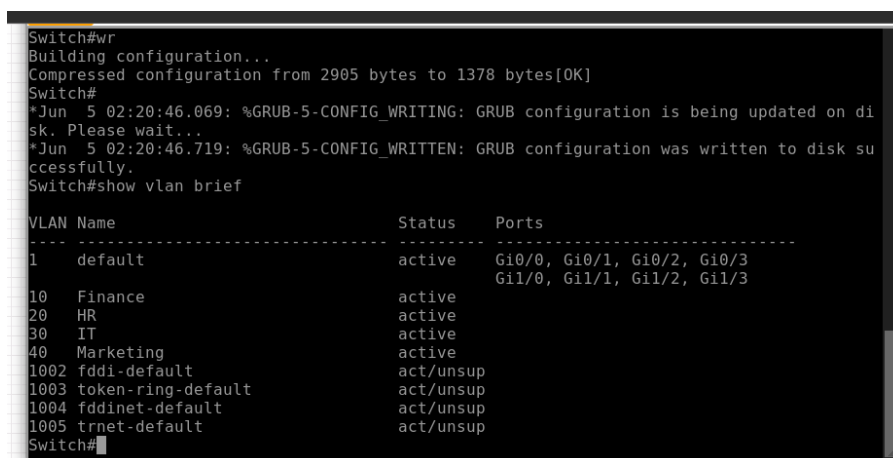
```

Gambar 2: Pembuatan Seluruh VLAN

3. Verifikasi Pembuatan VLAN

Memastikan seluruh VLAN database telah aktif.

```
show vlan brief
```



```

Switch#wr
Building configuration...
Compressed configuration from 2905 bytes to 1378 bytes[OK]
Switch#
*Jun  5 02:20:46.069: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on di
sk. Please wait...
*Jun  5 02:20:46.719: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk su
ccessfully.
Switch#show vlan brief

```

VLAN	Name	Status	Ports
1	default	active	Gi0/0, Gi0/1, Gi0/2, Gi0/3 Gi1/0, Gi1/1, Gi1/2, Gi1/3
10	Finance	active	
20	HR	active	
30	IT	active	
40	Marketingg	active	
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trnet-default	act/unsup	

```

Switch#

```

Gambar 3: Show VLAN Brief

4. Konfigurasi Port Akses (Access Port) Client

Memetakan interface fisik switch menuju ke VLAN ID client masing-masing.

```

Switch# configure terminal

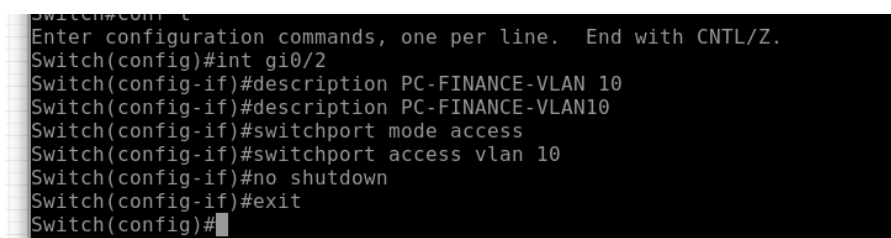
interface gi0/2
  description PC-FINANCE-VLAN10
  switchport mode access
  switchport access vlan 10
  no shutdown
exit

interface gi0/1
  description PC-HR-VLAN20
  switchport mode access
  switchport access vlan 20
  no shutdown
exit

interface gi1/1
  description PC-IT-VLAN30
  switchport mode access
  switchport access vlan 30
  no shutdown
exit

interface gi0/3
  description PC-MARKETING-VLAN40
  switchport mode access
  switchport access vlan 40
  no shutdown
exit

```

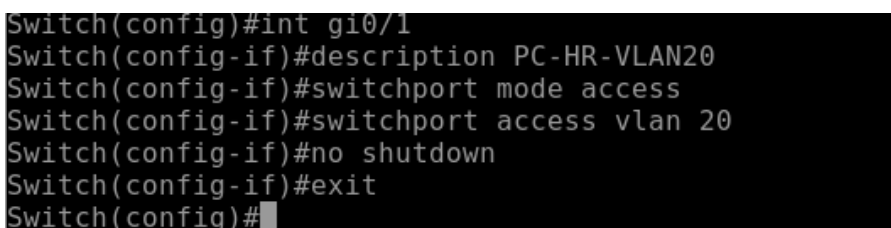


```

Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int gi0/2
Switch(config-if)#description PC-FINANCE-VLAN 10
Switch(config-if)#description PC-FINANCE-VLAN10
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#

```

Gambar 4: Konfigurasi Access Port Interface Gi0/2



```

Switch(config)#int gi0/1
Switch(config-if)#description PC-HR-VLAN20
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#

```

Gambar 5: Konfigurasi Access Port Interface Gi0/1

```
Switch(config)#int gi1/1
Switch(config-if)#description PC-IT-VLAN30
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#
```

Gambar 6: Konfigurasi Access Port Interface Gi1/1

```
Switch(config-if)#int gi0/3
Switch(config-if)#description PC-Marketing-VLAN40
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 40
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#
```

Gambar 7: Konfigurasi Access Port Interface Gi0/3

5. Konfigurasi Jalur Trunk ke Cisco Router

Mengatur port gi0/0 agar dapat melewatkan paket data ter-tagging dari VLAN 10, 20, 30, dan 40.

Switch# configure terminal

```
interface gi0/0
  switchport trunk encapsulation dot1q
  description TRUNK-TO-CISCO-ROUTER
  switchport mode trunk
  switchport trunk allowed vlan 10,20,30,40
  no shutdown
exit
```

```
Switch(config)#int gi0/0
Switch(config-if)#switchport trunk encapsulation dot1q
Switch(config-if)#description TRUNK-TO-CISCO-ROUTER
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk allowed vlan 10,20,30,40
Switch(config-if)#no shutdown
Switch(config-if)#exit
```

Gambar 8: Konfigurasi Jalur Trunk Port gi0/0

6. Verifikasi Status Trunking dan Penyimpanan Konfigurasi

Memeriksa status enkapsulasi trunk dan menyimpan pengaturan ke NVRAM.

```
show interfaces trunk
write memory
```

```

*Jun  5 02:25:20.832: %SYS-5-CONFIG_I: Configured from console by console
Switch#vr
Building configuration...
Compressed configuration from 3364 bytes to 1645 bytes[OK]
Switch#
*Jun  5 02:25:24.744: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on di
sk. Please wait...
*Jun  5 02:25:25.404: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk su
ccessfully.show interface trunk

Port      Mode      Encapsulation  Status      Native vlan
Gi0/0     on        802.1q         trunking    1

Port      Vlans allowed on trunk
Gi0/0     10,20,30,40

Port      Vlans allowed and active in management domain
Gi0/0     10,20,30,40

Port      Vlans in spanning tree forwarding state and not pruned
Gi0/0     10,20,30,40
Switch#

```

Gambar 9: Verifikasi Hasil show interfaces trunk

1.3 Praktikum 2: Konfigurasi Cisco Router

Cisco Router dikonfigurasi sebagai pusat *gateway* antar-VLAN dengan metode *Router-on-a-Stick* (ROAS) sekaligus bertindak sebagai DHCP Server.

1. Masuk ke Mode Konfigurasi Global Cisco Router

```
enable
configure terminal
```

2. Aktivasi Interface Fisik Trunk

Menghidupkan interface gig0/2 yang mengarah ke switch tanpa memberikan IP address utama.

```
interface gig0/2
description TRUNK-TO-CISCO-SWITCH
no ip address
no shutdown
exit
```

```

Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int gig0/2
Router(config-if)#description TRUNK-TO-CISCO-SWITCH
Router(config-if)#no ip address
Router(config-if)#no shutdown
Router(config-if)#exit

```

Gambar 10: Aktivasi dan Konfigurasi Interface Trunk gig0/2

3. Konfigurasi Subinterface VLAN 10 (Finance)

```
interface gig0/2.10
description GATEWAY-VLAN10-FINANCE
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
no shutdown
exit
```

```

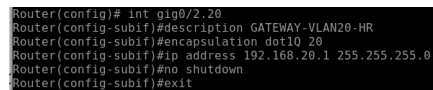
Router(config)#int gig0/2.10
Router(config-subif)#description GATEWAY-VLAN10-FINANCE
Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit

```

Gambar 11: Konfigurasi Subinterface gig0/2.10

4. Konfigurasi Subinterface VLAN 20 (HR)

```
interface gig0/2.20
description GATEWAY-VLAN20-HR
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
no shutdown
exit
```

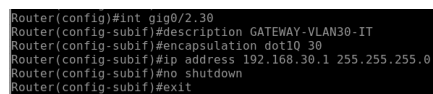


```
Router(config)# int gig0/2.20
Router(config-subif)#description GATEWAY-VLAN20-HR
Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
```

Gambar 12: Konfigurasi Subinterface gig0/2.20

5. Konfigurasi Subinterface VLAN 30 (IT)

```
interface gig0/2.30
description GATEWAY-VLAN30-IT
encapsulation dot1Q 30
ip address 192.168.30.1 255.255.255.0
no shutdown
exit
```

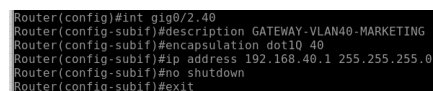


```
Router(config)#int gig0/2.30
Router(config-subif)#description GATEWAY-VLAN30-IT
Router(config-subif)#encapsulation dot1Q 30
Router(config-subif)#ip address 192.168.30.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
```

Gambar 13: Konfigurasi Subinterface gig0/2.30

6. Konfigurasi Subinterface VLAN 40 (Marketing)

```
interface gig0/2.40
description GATEWAY-VLAN40-MARKETING
encapsulation dot1Q 40
ip address 192.168.40.1 255.255.255.0
no shutdown
exit
```



```
Router(config)#int gig0/2.40
Router(config-subif)#description GATEWAY-VLAN40-MARKETING
Router(config-subif)#encapsulation dot1Q 40
Router(config-subif)#ip address 192.168.40.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
```

Gambar 14: Konfigurasi Subinterface gig0/2.40

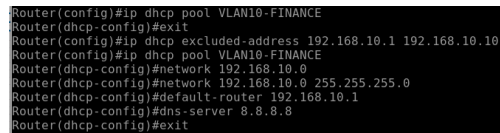
7. Konfigurasi DHCP Pool untuk VLAN 10

Mengecualikan rentang IP 192.168.10.1 hingga 192.168.10.10, lalu membuat pool DHCP.

```
ip dhcp excluded-address 192.168.10.1 192.168.10.10
```



```
ip dhcp pool VLAN10-FINANCE
network 192.168.10.0 255.255.255.0
default-router 192.168.10.1
dns-server 8.8.8.8
exit
```



```
Router(config)#ip dhcp pool VLAN10-FINANCE
Router(dhcp-config)#exit
Router(config)#ip dhcp excluded-address 192.168.10.1 192.168.10.10
Router(config)#ip dhcp pool VLAN10-FINANCE
Router(dhcp-config)#network 192.168.10.0
Router(dhcp-config)#network 192.168.10.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.10.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
```

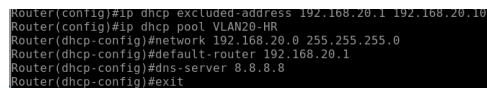
Gambar 15: Konfigurasi DHCP Server untuk VLAN 10

8. Konfigurasi DHCP Pool untuk VLAN 20

Mengecualikan rentang IP 192.168.20.1 hingga 192.168.20.10, lalu membuat pool DHCP.

```
ip dhcp excluded-address 192.168.20.1 192.168.20.10

ip dhcp pool VLAN20-HR
network 192.168.20.0 255.255.255.0
default-router 192.168.20.1
dns-server 8.8.8.8
exit
```



```
Router(config)#ip dhcp excluded-address 192.168.20.1 192.168.20.10
Router(config)#ip dhcp pool VLAN20-HR
Router(dhcp-config)#network 192.168.20.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.20.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
```

Gambar 16: Konfigurasi DHCP Server untuk VLAN 20

9. Konfigurasi Alokasi IP Statis dan Dinamis pada VPCS

Pemberian parameter jaringan secara manual pada PC VLAN 30 and 40, serta otomatis untuk PC VLAN 10 dan 20.

```
PC-IT> ip 192.168.30.10/24 192.168.30.1
PC-Marketing> ip 192.168.40.10/24 192.168.40.1
PC-Finance> ip dhcp
PC-HR> ip dhcp
```

10. Verifikasi Status Antarmuka Jaringan dan DHCP Pool

Melakukan pemeriksaan ringkas status IP *subinterface*, pool DHCP, dan pemetaan client aktif di router.

```
show ip interface brief
show ip dhcp pool
show ip dhcp binding
```

```

Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#show ip interface brief

```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	unassigned	YES	unset	administratively down	down
GigabitEthernet0/1	unassigned	YES	unset	administratively down	down
GigabitEthernet0/2	unassigned	YES	unset	up	up
GigabitEthernet0/2.10	192.168.10.1	YES	manual	up	up
GigabitEthernet0/2.20	192.168.20.1	YES	manual	up	up
GigabitEthernet0/2.30	192.168.30.1	YES	manual	up	up
GigabitEthernet0/2.40	192.168.40.1	YES	manual	up	up
GigabitEthernet0/3	unassigned	YES	unset	administratively down	down

```

Router#

```

Gambar 17: Output Ringkas Status Antarmuka Jaringan

```

GigabitEthernet0/3      unassigned      YES unset      administratively down down
Router#show ip dhcp pool

```

```

Pool VLAN10-FINANCE :
  Utilization mark (high/low) : 100 / 0
  Subnet size (first/next) : 0 / 0
  Total addresses : 254
  Leased addresses : 1
  Pending event : none
  1 subnet is currently in the pool :
    Current index   IP address range   Leased addresses
    192.168.10.12   192.168.10.1 - 192.168.10.254   1

```

```

Pool VLAN20-HR :
  Utilization mark (high/low) : 100 / 0
  Subnet size (first/next) : 0 / 0
  Total addresses : 254
  Leased addresses : 1
  Pending event : none
  1 subnet is currently in the pool :
    Current index   IP address range   Leased addresses
    192.168.20.12   192.168.20.1 - 192.168.20.254   1
Router#

```

Gambar 18: Output Status DHCP Pool

```

Router#show ip dhcp binding

```

IP address	Client-ID/ Hardware address/ User name	Lease expiration	Type
192.168.10.11	0100.5079.6668.3f	Jun 06 2026 02:40 AM	Automatic
192.168.20.11	0100.5079.6668.40	Jun 06 2026 02:39 AM	Automatic

```

Router#

```

Gambar 19: Output Hasil show ip dhcp binding

11. Pengujian Verifikasi Koneksi VPCS Lokal

Melakukan pengecekan perolehan IP menggunakan `show ip`, dilanjutkan dengan *ping* ke masing-masing *gateway*.

PC-Finance> show ip

PC-Finance> ping 192.168.10.1

PC-HR> show ip

PC-HR> ping 192.168.20.1

PC-IT> ping 192.168.30.1

PC-Marketing> ping 192.168.40.1

```

VPCS> show ip

```

NAME	: VPCS[1]
IP/MASK	: 192.168.10.11/24
GATEWAY	: 192.168.10.1
DNS	: 8.8.8.8
DHCP SERVER	: 192.168.10.1
DHCP LEASE	: 86396, 86400/43200/75600
MAC	: 00:50:79:66:68:3f
LPORT	: 20000
RHOST:PORT	: 127.0.0.1:30000
MTU	: 1500

Gambar 20: Pengecekan Alokasi IP PC-Finance

```

VPCS>
VPCS> ping 192.168.10.1
84 bytes from 192.168.10.1 icmp_seq=1 ttl=255 time=1.778 ms
84 bytes from 192.168.10.1 icmp_seq=2 ttl=255 time=2.479 ms
84 bytes from 192.168.10.1 icmp_seq=3 ttl=255 time=2.283 ms
84 bytes from 192.168.10.1 icmp_seq=4 ttl=255 time=2.732 ms
84 bytes from 192.168.10.1 icmp_seq=5 ttl=255 time=3.017 ms
VPCS>

```

Gambar 21: Uji Ping Gateway dari PC-Finance

```

VPCS> ip dhcp
DDORA IP 192.168.20.11/24 GW 192.168.20.1
VPCS> show ip
NAME      : VPCS[1]
IP/MASK   : 192.168.20.11/24
GATEWAY   : 192.168.20.1
DNS       : 8.8.8.8
DHCP SERVER : 192.168.20.1
DHCP LEASE : 86163, 86480/43200/75600
MAC       : 00:50:79:66:60:40
PORT      : 20000
RHOST:PORT : 127.0.0.1:30000
MTU       : 1500
VPCS>

```

Gambar 22: Pengecekan Alokasi IP PC-HR

```

VPCS> ping 192.168.20.1
84 bytes from 192.168.20.1 icmp_seq=1 ttl=255 time=1.533 ms
84 bytes from 192.168.20.1 icmp_seq=2 ttl=255 time=4.361 ms
84 bytes from 192.168.20.1 icmp_seq=3 ttl=255 time=2.658 ms
84 bytes from 192.168.20.1 icmp_seq=4 ttl=255 time=2.375 ms

```

Gambar 23: Uji Ping Gateway dari PC-HR

```

VPCS> ping 192.168.30.1
84 bytes from 192.168.30.1 icmp_seq=1 ttl=255 time=2.074 ms
84 bytes from 192.168.30.1 icmp_seq=2 ttl=255 time=3.526 ms
84 bytes from 192.168.30.1 icmp_seq=3 ttl=255 time=3.398 ms
84 bytes from 192.168.30.1 icmp_seq=4 ttl=255 time=4.063 ms

```

Gambar 24: Uji Ping Gateway dari PC-IT

```

VPCS> ping 192.168.40.1
84 bytes from 192.168.40.1 icmp_seq=1 ttl=255 time=2.392 ms
84 bytes from 192.168.40.1 icmp_seq=2 ttl=255 time=1.765 ms
84 bytes from 192.168.40.1 icmp_seq=3 ttl=255 time=2.671 ms
84 bytes from 192.168.40.1 icmp_seq=4 ttl=255 time=1.861 ms
84 bytes from 192.168.40.1 icmp_seq=5 ttl=255 time=2.528 ms

```

Gambar 25: Uji Ping Gateway dari PC-Marketing

12. Pengujian Konektivitas Lintas Antar-VLAN

Menguji kemampuan *Inter-VLAN routing* dari PC Finance (VLAN 10) menuju client di segmen VLAN lain.

PC-Finance> ping 192.168.20.11

PC-Finance> ping 192.168.30.10

PC-Finance> ping 192.168.40.10

```

VPCS> ping 192.168.20.11
192.168.20.11 icmp_seq=1 ttl=64 time=0.001 ms
192.168.20.11 icmp_seq=2 ttl=64 time=0.001 ms
192.168.20.11 icmp_seq=3 ttl=64 time=0.001 ms
192.168.20.11 icmp_seq=4 ttl=64 time=0.001 ms
192.168.20.11 icmp_seq=5 ttl=64 time=0.001 ms

VPCS> ping 192.168.30.10
64 bytes from 192.168.30.10 icmp_seq=1 ttl=63 time=7.146 ms
64 bytes from 192.168.30.10 icmp_seq=2 ttl=63 time=3.105 ms
64 bytes from 192.168.30.10 icmp_seq=3 ttl=63 time=2.675 ms
64 bytes from 192.168.30.10 icmp_seq=4 ttl=63 time=2.266 ms
64 bytes from 192.168.30.10 icmp_seq=5 ttl=63 time=2.611 ms

VPCS> ping 192.168.40.10
64 bytes from 192.168.40.10 icmp_seq=1 ttl=63 time=5.050 ms
64 bytes from 192.168.40.10 icmp_seq=2 ttl=63 time=2.260 ms
64 bytes from 192.168.40.10 icmp_seq=3 ttl=63 time=2.488 ms
64 bytes from 192.168.40.10 icmp_seq=4 ttl=63 time=2.730 ms

```

Gambar 26: Hasil Uji Uji Konektivitas Antar-VLAN dari PC-Finance

1.4 Praktikum 3: Konfigurasi Huawei Router dan IP Address Antarrouter

Konfigurasi diarahkan untuk membangun segmen lokal baru pada infrastruktur Huawei Router serta pengalokasian alamat IP *point-to-point* pada tautan penghubung antarrouter.

1. Konfigurasi Alamat IP LAN Huawei

```

system-view
interface Ethernet1/0/2
description LAN-HUAWEI
ip address 192.168.50.1 255.255.255.0
undo shutdown
quit
commit

```

```

Enter system view, return user view with return command.
[~HUAWEI]
[~HUAWEI]interface Ethernet1/0/2
[~HUAWEI-Ethernet1/0/2] description LAN-HUAWEI
[*HUAWEI-Ethernet1/0/2] ip address 192.168.50.1 255.255.255.0
[*HUAWEI-Ethernet1/0/2] undo shutdown
[*HUAWEI-Ethernet1/0/2] quit
[*HUAWEI]
[*HUAWEI]commit
[~HUAWEI]

```

Gambar 27: Konfigurasi Interface LAN Ethernet1/0/2 Sisi Huawei

2. Konfigurasi IP VPCS LAN Huawei dan Pengujian Jaringan

Menentukan konfigurasi statis client dan memverifikasi respon tautan ke *gateway* Huawei.

```

PC-Huawei> ip 192.168.50.10/24 192.168.50.1
PC-Huawei> show ip
PC-Huawei> ping 192.168.50.1

```

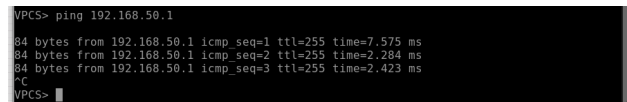
```

VPCS> ip 192.168.50.10/24 192.168.50.1
Checking for duplicate address...
PC1 : 192.168.50.10 255.255.255.0 gateway 192.168.50.1

VPCS> show ip
NAME       : VPCS[1]
IP/MASK    : 192.168.50.10/24
GATEWAY    : 192.168.50.1
DNS        :
MAC        : 00:50:79:66:68:3e
LPORT      : 20000
RHOST:PORT : 127.0.0.1:30000
MTU        : 1500

```

Gambar 28: Verifikasi Alokasi IP PC LAN Huawei



```

VPCS> ping 192.168.50.1
84 bytes from 192.168.50.1 icmp_seq=1 ttl=255 time=7.575 ms
84 bytes from 192.168.50.1 icmp_seq=2 ttl=255 time=2.284 ms
84 bytes from 192.168.50.1 icmp_seq=3 ttl=255 time=2.423 ms
^C
VPCS>

```

Gambar 29: Pengujian Koneksi Ping Menuju Gateway Huawei

3. Konfigurasi Alamat IP Jalur Cisco ↔ Huawei Link 1

Menggunakan alokasi network 10.10.10.0/30.

! Di Sisi Cisco Router

```

interface gi0/3
description LINK1-TO-HUAWEI
ip address 10.10.10.1 255.255.255.252
no shutdown
exit

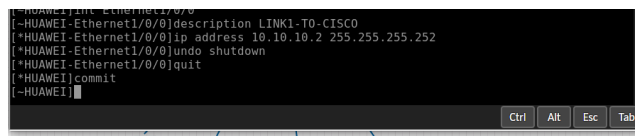
```

! Di Sisi Huawei Router

```

system-view
interface Ethernet1/0/0
description LINK1-TO-CISCO
ip address 10.10.10.2 255.255.255.252
undo shutdown
quit
commit

```



```

[*HUAWEI]int Ethernet1/0/0
[*HUAWEI-Ethernet1/0/0]description LINK1-TO-CISCO
[*HUAWEI-Ethernet1/0/0]ip address 10.10.10.2 255.255.255.252
[*HUAWEI-Ethernet1/0/0]undo shutdown
[*HUAWEI-Ethernet1/0/0]quit
[*HUAWEI]commit
[*HUAWEI]

```

Gambar 30: Konfigurasi IP Link 1 Terhubung ke Cisco Pada Huawei Router

4. Konfigurasi Alamat IP Jalur Cisco ↔ Huawei Link 2

Menggunakan alokasi network 10.10.11.0/30.

! Di Sisi Cisco Router

```

interface gi0/0
description LINK2-TO-HUAWEI
ip address 10.10.11.1 255.255.255.252
no shutdown
exit

```

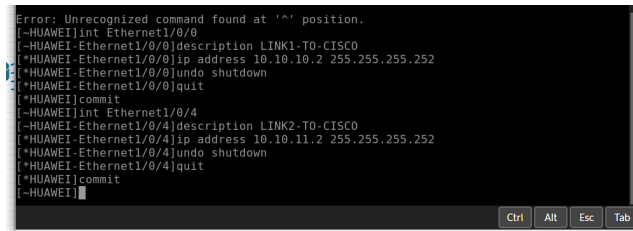
! Di Sisi Huawei Router

```

system-view
interface Ethernet1/0/4
description LINK2-TO-CISCO
ip address 10.10.11.2 255.255.255.252
undo shutdown

```

```
quit
commit
```



Gambar 31: Konfigurasi IP Link 2 Terhubung ke Cisco Pada Huawei Router

5. Konfigurasi Alamat IP Jalur Huawei ↔ MikroTik

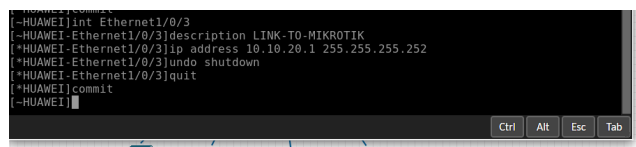
Menggunakan alokasi network 10.10.20.0/30.

! Di Sisi Huawei Router

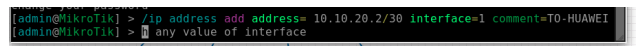
```
system-view
interface Ethernet1/0/3
description LINK-TO-MIKROTIK
ip address 10.10.20.1 255.255.255.252
undo shutdown
quit
commit
```

Di Sisi MikroTik

```
/ip address add address=10.10.20.2/30 interface=ether1 comment=TO-HUAWEI
```



Gambar 32: Konfigurasi IP Link Huawei ke MikroTik



Gambar 33: Konfigurasi IP Alamat Link di MikroTik Sisi ether1

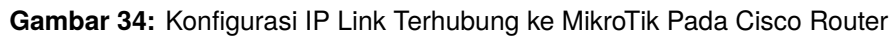
6. Konfigurasi Alamat IP Jalur Cisco ↔ MikroTik

Menggunakan alokasi network 10.10.30.0/30.

! Di Sisi Cisco Router

```
interface gi0/1
description LINK-TO-MIKROTIK
ip address 10.10.30.1 255.255.255.252
no shutdown
exit
```

```
/ip address add address=10.10.30.2/30 interface=ether2 comment=T0-CISCO
```



Memastikan status fisik maupun logis dari antarmuka inter-router bernilai aktif (*up*).

```
show ip interface brief
```

```
display ip interface brief
```

```
/ip address print
```



```
admin@MikroTik:~$ ip address print
Flags: X - disabled, I - invalid, D - dynamic
# ADDRESS NETWORK INTERFACE
0 :::: TO-CISCO
1 10.10.30.2/30 10.10.30.0 ether2
2 10.10.20.2/30 10.10.20.0 ether1
admin@MikroTik:~$
```

Gambar 38: Hasil Cetak Address Print Sisi MikroTik

8. Pengujian Validasi Konektivitas Tetangga Langsung (Directly Connected)

Mengeksekusi perintah *ping* titik-ke-titik guna memastikan respon antar-antarmuka berjalan sebelum OSPF diaktifkan.

! Pengujian dari Cisco Router

ping 10.10.10.2

ping 10.10.11.2

ping 10.10.30.2

! Pengujian dari Huawei Router

ping 10.10.10.1

ping 10.10.11.1

ping 10.10.20.2

Pengujian dari MikroTik Router

ping 10.10.20.1

ping 10.10.30.1

```
Router#ping 10.10.10.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/3 ms
Router#ping 10.10.11.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.11.2, timeout is 2 seconds:
!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/3 ms
Router#ping 10.10.30.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.30.2, timeout is 2 seconds:
!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/2 ms
Router#
```

Gambar 39: Sukses Pengujian Ping Tetangga Langsung Dari Cisco Router

```
[~HUAWEI]ping 10.10.10.1
PING 10.10.10.1: 56 data bytes, press CTRL C to break
Reply from 10.10.10.1: bytes=56 Sequence=1 ttl=255 time=4 ms
Reply from 10.10.10.1: bytes=56 Sequence=2 ttl=255 time=2 ms
Reply from 10.10.10.1: bytes=56 Sequence=3 ttl=255 time=2 ms
Reply from 10.10.10.1: bytes=56 Sequence=4 ttl=255 time=2 ms
Reply from 10.10.10.1: bytes=56 Sequence=5 ttl=255 time=2 ms

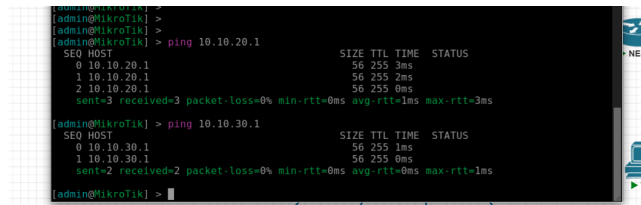
--- 10.10.10.1 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 2/2/4 ms

[~HUAWEI]ping 10.10.11.1
PING 10.10.11.1: 56 data bytes, press CTRL C to break
Reply from 10.10.11.1: bytes=56 Sequence=1 ttl=255 time=1 ms
Reply from 10.10.11.1: bytes=56 Sequence=2 ttl=255 time=1 ms
Reply from 10.10.11.1: bytes=56 Sequence=3 ttl=255 time=1 ms
Reply from 10.10.11.1: bytes=56 Sequence=4 ttl=255 time=1 ms

--- 10.10.11.1 ping statistics ---
4 packet(s) transmitted
4 packet(s) received
0.00% packet loss
round-trip min/avg/max = 1/1/2 ms

[~HUAWEI]
```

Gambar 40: Sukses Pengujian Ping Tetangga Langsung Dari Huawei Router



Gambar 41: Sukses Pengujian Ping Tetangga Langsung Dari MikroTik

1.5 Praktikum 4: Konfigurasi OSPF MultiVendor dan OSPF Cost

Protokol *routing* dinamis OSPF Area 0 diaktifkan untuk mendistribusikan tabel *routing* multi-vendor secara otomatis, dengan memanipulasi nilai *cost* untuk menentukan prioritas jalur utama dan cadangan.

Tautan Penghubung (Link)	Nilai OSPF Cost	Penentuan Peran Jalur
Cisco ↔ Huawei Link 1	10	Jalur Utama (<i>Primary</i>)
Cisco ↔ Huawei Link 2	10	Jalur Utama (<i>Primary</i>)
Cisco ↔ MikroTik	100	Jalur Cadangan (<i>Backup</i>)
Huawei ↔ MikroTik	100	Jalur Cadangan (<i>Backup</i>)

1. Konfigurasi OSPF and OSPF Cost pada Cisco Router

Mengaktifkan proses OSPF ID 1, mendefinisikan *Router ID*, mendaftarkan 7 segmen *network*, dan menerapkan nilai *cost*.

```

router ospf 1
  router-id 1.1.1.1
  network 10.10.10.0 0.0.0.3 area 0
  network 10.10.11.0 0.0.0.3 area 0
  network 10.10.30.0 0.0.0.3 area 0
  network 192.168.10.0 0.0.0.255 area 0
  network 192.168.20.0 0.0.0.255 area 0
  network 192.168.30.0 0.0.0.255 area 0
  network 192.168.40.0 0.0.0.255 area 0
  exit

```

```

interface gi0/3
  ip ospf cost 10
  exit
interface gi0/0
  ip ospf cost 10
  exit
interface gi0/1
  ip ospf cost 100
  exit

```

```

Router(config)#router ospf 1
Router(config-router)#router-id 1.1.1.1
Router(config-router)#network 10.10.10.0 0.0.0.3 area 0
Router(config-router)#network 10.10.11.0 0.0.0.3 area 0
Router(config-router)#network 10.10.30.0 0.0.0.3 area 0
Router(config-router)#network 192.168.10.0 0.0.0.255 area 0
Router(config-router)#network 192.168.20.0 0.0.0.255 area 0
Router(config-router)#network 192.168.30.0 0.0.0.255 area 0
Router(config-router)#network 192.168.40.0 0.0.0.255 area 0
Router(config-router)#exit
Router(config)#

```

Gambar 42: Pernyataan Berbagi Segmen Jaringan OSPF di Cisco

```

Router(config)#interface gi0/3
Router(config-if)#description LINK1-TO-HUAWEI
Router(config-if)#ip ospf cost 10
Router(config-if)#exit
Router(config)#interface gi0/0
Router(config-if)#description LINK2-TO-HUAWEI
Router(config-if)#ip ospf cost 10
Router(config-if)#exit
Router(config)#interface gi0/1
Router(config-if)#description LINK-TO-MIKROTIK
Router(config-if)#ip ospf cost 100
Router(config-if)#exit

```

Gambar 43: Pengaturan Modifikasi Parameter Cost di Antarmuka Cisco

2. Verifikasi Awal di Cisco Router

```

show ip ospf neighbor
show ip route ospf
show ip protocols

```

```

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.10.10.0 0.0.0.3 area 0
    10.10.11.0 0.0.0.3 area 0
    10.10.30.0 0.0.0.3 area 0
    192.168.10.0 0.0.0.255 area 0
    192.168.20.0 0.0.0.255 area 0
    192.168.30.0 0.0.0.255 area 0
    192.168.40.0 0.0.0.255 area 0
  Routing Information Sources:
    Gateway         Distance         Last Update
  Distance: (default is 110)

```

Gambar 44: Verifikasi Protokol Informasi Aktif Melalui show ip protocols

3. Konfigurasi OSPF and OSPF Cost pada Huawei Router

Mengaktifkan proses OSPF ID 1, mendefinisikan *Router ID*, mendaftarkan 4 segmen *network*, dan menetapkan parameter *cost*.

```

system-view
ospf 1 router-id 2.2.2.2
area 0.0.0.0
  network 10.10.10.0 0.0.0.3
  network 10.10.11.0 0.0.0.3
  network 10.10.20.0 0.0.0.3
  network 192.168.50.0 0.0.0.255
quit
quit

interface Ethernet1/0/0
  ospf cost 10
quit

```

```

interface Ethernet1/0/4
  ospf cost 10
quit
interface Ethernet1/0/3
  ospf cost 100
quit
commit

```

```

[~HUAWEI]ospf 1 router-id 2.2.2.2
[*HUAWEI-ospf-1]area 0.0.0.0
[*HUAWEI-ospf-1-area-0.0.0.0]network 10.10.10.0 0.0.0.3
[*HUAWEI-ospf-1-area-0.0.0.0]network 10.10.11.0 0.0.0.3
[*HUAWEI-ospf-1-area-0.0.0.0]network 10.10.20.0 0.0.0.3
[*HUAWEI-ospf-1-area-0.0.0.0]network 192.168.50.0 0.0.0.255
[*HUAWEI-ospf-1-area-0.0.0.0]quit
[*HUAWEI-ospf-1]quit

```

Gambar 45: Konfigurasi Penambahan OSPF ID dan Area Backbone Sisi Huawei

```

[*HUAWEI]interface Ethernet1/0/0
[*HUAWEI-Ethernet1/0/0]description LINK1-T0-CISCO
[*HUAWEI-Ethernet1/0/0]ospf cost10
^
Error: Unrecognized command found at '^' position.
[*HUAWEI-Ethernet1/0/0]ospf cost 10
[*HUAWEI-Ethernet1/0/0]quit
[*HUAWEI]interface Ethernet1/0/4
[*HUAWEI-Ethernet1/0/4]description LINK2-T0-CISCO
[*HUAWEI-Ethernet1/0/4]ospf cost 10
[*HUAWEI-Ethernet1/0/4]quit
[*HUAWEI]interface Ethernet1/0/3
[*HUAWEI-Ethernet1/0/3]
Warning: Uncommitted configurations found, commit them before exiting? [Y(yes)/N(no)/C(cancel)]):C
[*HUAWEI-Ethernet1/0/3]description LINK-T0-MIKROTIK
[*HUAWEI-Ethernet1/0/3]ospf cost 100
[*HUAWEI-Ethernet1/0/3]quit
[*HUAWEI]commit

```

Gambar 46: Penerapan Modifikasi Cost di Antarmuka Huawei

4. Verifikasi Kedekatan dan Informasi Rute OSPF pada Huawei

```

display ospf peer
display ip routing-table protocol ospf
display ospf routing

```

```

[~HUAWEI]display ospf peer
(M) Indicates MADJ neighbor

      OSPF Process 1 with Router ID 2.2.2.2
      Neighbors

Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
Router ID: 1.1.1.1      Address: 10.10.10.1
State: Full           Mode:Nbr is Slave   Priority: 1
DR: 10.10.10.1        BDR: 10.10.10.2     MTU: 1500
Dead timer due in 30 sec
Retrans timer interval: 5
Neighbor is up for 00h00m31s
Neighbor Up Time : 2026-06-05 03:47:04
Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
Router ID: 1.1.1.1      Address: 10.10.11.1
State: Full           Mode:Nbr is Slave   Priority: 1
DR: 10.10.11.1        BDR: 10.10.11.2     MTU: 1500
Dead timer due in 34 sec
Retrans timer interval: 5
Neighbor is up for 00h00m31s
Neighbor Up Time : 2026-06-05 03:47:04
Authentication Sequence: [ 0 ]

[~HUAWEI]

```

Gambar 47: Pengecekan Status Kedekatan Melalui display ospf peer

```

[~HUAWEI]display ip routing-table protocol ospf
Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route
-----
public Routing Table : OSPF
Destinations : 9          Routes : 14

OSPF routing table status : <Active>
Destinations : 5          Routes : 10

Destination/Mask    Proto    Pre    Cost    Flags NextHop         Interface
-----
10.10.30.0/30       OSPF     10     110     D    10.10.11.1            Ethernet1/0/4
10.10.30.0/30       OSPF     10     110     D    10.10.10.1            Ethernet1/0/0
192.168.10.0/24     OSPF     10     11       D    10.10.11.1            Ethernet1/0/4
192.168.20.0/24     OSPF     10     11       D    10.10.10.1            Ethernet1/0/0
192.168.30.0/24     OSPF     10     11       D    10.10.11.1            Ethernet1/0/4
192.168.40.0/24     OSPF     10     11       D    10.10.10.1            Ethernet1/0/0
192.168.40.0/24     OSPF     10     11       D    10.10.11.1            Ethernet1/0/4
192.168.40.0/24     OSPF     10     11       D    10.10.10.1            Ethernet1/0/0

OSPF routing table status : <Inactive>
Destinations : 4          Routes : 4

Destination/Mask    Proto    Pre    Cost    Flags NextHop         Interface
-----
10.10.10.0/30       OSPF     10     10       D    10.10.10.2            Ethernet1/0/0
10.10.11.0/30       OSPF     10     10       D    10.10.11.2            Ethernet1/0/4

```

Gambar 48: Informasi Tabel Rute Eksternal OSPF pada Huawei

```

(M) Indicates MADJ neighbor

OSPF Process 1 with Router ID 2.2.2.2
Neighbors

Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
Router ID: 1.1.1.1      Address: 10.10.10.1
State: Full             Mode:Nbr is Slave      Priority: 1
DR: 10.10.10.1          BDR: 10.10.10.2      MTU: 1500
Dead timer due in 35 sec
Retrans timer interval: 5
Neighbor is up for 00h08m08s
Neighbor Up Time : 2026-06-05 03:47:04
Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.20.1 (Eth1/0/3)'s neighbors
Router ID: 3.3.3.3      Address: 10.10.20.2
State: Full             Mode:Nbr is Master     Priority: 1
DR: 10.10.20.1          BDR: 10.10.20.2      MTU: 1500
Dead timer due in 32 sec
Retrans timer interval: 5
Neighbor is up for 00h01m51s

```

Gambar 49: Hasil Output Detail display ospf routing

5. Konfigurasi OSPF and OSPF Cost pada MikroTik

MikroTik dikonfigurasi sebagai node penyimpanan rute cadangan (*backup*).

```
/routing ospf instance set default router-id=3.3.3.3
```

```
/routing ospf network add network=10.10.20.0/30 area=backbone
```

```
/routing ospf network add network=10.10.30.0/30 area=backbone
```

```
/routing ospf interface add interface=ether1 cost=100
```

```
/routing ospf interface add interface=ether2 cost=100
```

```

[admin@MikroTik] > /routing ospf neighbor print
0 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 prio
rity=1
dr-address=10.10.30.1 backup-dr-address=0.0.0.0 state="Full" state-changes=
5
ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=9s
1 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 prio
rity=1
dr-address=10.10.20.1 backup-dr-address=10.10.20.2 state="Full" state-chang
es=5
ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=23s
[admin@MikroTik] >

```

Gambar 50: Pernyataan Alokasi Node Instance ID dan Network pada MikroTik

```
[admin@MikroTik] > /ip route print where ospf
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
#   DST-ADDRESS      PREF-SRC  GATEWAY      DISTANCE
0   ADo 10.10.10.0/30          10.10.30.1      110
    10.10.20.1
1   ADo 10.10.11.0/30          10.10.30.1      110
    10.10.20.1
2   ADo 192.168.10.0/24         10.10.30.1      110
3   ADo 192.168.20.0/24         10.10.30.1      110
4   ADo 192.168.30.0/24         10.10.30.1      110
5   ADo 192.168.40.0/24         10.10.30.1      110
6   ADo 192.168.50.0/24         10.10.20.1      110
[admin@MikroTik] >
```

Gambar 51: Pemeriksaan Awal Tabel Rute OSPF Terbaca Sisi MikroTik

6. Verifikasi Status Relasi Tetangga (OSPF Neighbor) Global

Memastikan hubungan ketetanggaan antar-perangkat telah terbentuk secara menyeluruh.

! Di Cisco (Harus mendeteksi 2 Link Huawei dan 1 Link MikroTik)

show ip ospf neighbor

! Di Huawei (Harus mendeteksi 2 Link Cisco dan 1 Link MikroTik)

display ospf peer

Di MikroTik (Harus mendeteksi Neighbor ID Cisco dan Huawei)

/routing ospf neighbor print

```
[admin@MikroTik] > /routing ospf neighbor print
0 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1
  dr-address=10.10.30.1 backup-dr-address=10.10.30.2 state="Full" state-changes=5
  ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=2m26s
1 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1
  dr-address=10.10.20.1 backup-dr-address=10.10.20.2 state="Full" state-changes=5
  ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=2m40s
[admin@MikroTik] >
```

Gambar 52: Verifikasi Kedekatan Global Cetakan Sisi MikroTik

```
Area 0.0.0.0 interface 10.10.20.1 (Eth1/0/3)'s neighbors
Router ID: 3.3.3.3 Address: 10.10.20.2
State: Full Mode:Nbr is Master Priority: 1
DR: 10.10.20.1 BDR: 10.10.20.2 MTU: 1500
Dead timer due in 32 sec
Retrans timer interval: 5
Neighbor is up for 00h01m51s
Neighbor Up Time : 2026-06-05 03:53:21
Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
Router ID: 1.1.1.1 Address: 10.10.11.1
State: Full Mode:Nbr is Slave Priority: 1
DR: 10.10.11.1 BDR: 10.10.11.2 MTU: 1500
Dead timer due in 33 sec
Retrans timer interval: 5
Neighbor is up for 00h08m08s
Neighbor Up Time : 2026-06-05 03:47:04
Authentication Sequence: [ 0 ]

[~HUAWEI]
[~HUAWEI]
```

Gambar 53: Verifikasi Kedekatan Hubungan OSPF Sisi Huawei Ke Tetangga

7. Verifikasi Tabel Sinkronisasi Routing Jaringan

! Di Cisco (Verifikasi rute menuju LAN Huawei 192.168.50.0)

show ip route 192.168.50.0

Di MikroTik (Verifikasi tabel rute OSPF dinamis yang masuk)

/ip route print where ospf

```

Router#show ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        I - IS-IS, su - IS-IS summary, LL - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
        a - application route
        + - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
O    10.10.20.0/30 [110/110] via 10.10.11.2, 00:06:17, GigabitEthernet0/0
      [110/110] via 10.10.10.2, 00:06:17, GigabitEthernet0/3
O    192.168.50.0/24 [110/111] via 10.10.11.2, 00:06:17, GigabitEthernet0/0
      [110/111] via 10.10.10.2, 00:06:17, GigabitEthernet0/3
Router#

```

Gambar 54: Verifikasi Hasil Cetak show ip route ospf Sisi Cisco Router

```

[admin@MikroTik] > ip route print where ospf
Flags: X - disabled, A - active, D - dynamic,
        C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
        B - blackhole, U - unreachable, P - prohibit
#  DST-ADDRESS      PREF-SRC  GATEWAY            DISTANCE
0  Ado 10.10.10.0/30          10.10.30.1         110
1  Ado 10.10.11.0/30         10.10.20.1         110
2  Ado 192.168.10.0/24        10.10.30.1         110
3  Ado 192.168.20.0/24        10.10.30.1         110
4  Ado 192.168.30.0/24        10.10.30.1         110
5  Ado 192.168.40.0/24        10.10.30.1         110
6  Ado 192.168.50.0/24        10.10.20.1         110
[admin@MikroTik] >

```

Gambar 55: Verifikasi Sinkronisasi Rute OSPF Secara Lengkap Sisi MikroTik

8. Pengujian Koneksi Lintas End-to-End Jaringan (PC Client)

Mengeksekusi proses pengujian *ping* lintas domain router dari segmen VLAN Cisco menuju LAN Huawei, dan sebaliknya.

```

PC-Finance-V10> ping 192.168.50.10
PC-HR-V20> ping 192.168.50.10
PC-IT-V30> ping 192.168.50.10
PC-Marketing-V40> ping 192.168.50.10

```

```

PC-LAN-Huawei> ping 192.168.10.11
PC-LAN-Huawei> ping 192.168.20.11
PC-LAN-Huawei> ping 192.168.30.10
PC-LAN-Huawei> ping 192.168.40.10

```

```

VPCS> ping 192.168.50.10

84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=6.362 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=2.623 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=2.667 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=3.965 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=4.101 ms

VPCS>

```

Gambar 56: Sukses Pengujian Koneksi End-to-End Gambar 1

```

VPCS> ping 192.168.50.10

84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=3.165 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=3.912 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=3.586 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=4.995 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=5.210 ms

VPCS>
VPCS>

```

Gambar 57: Sukses Pengujian Koneksi End-to-End Gambar 2

```
VPCS> ping 192.168.50.10
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=3.374 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=2.811 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=4.430 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=2.585 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=3.448 ms
```

Gambar 58: Sukses Pengujian Koneksi End-to-End Gambar 3

```
VPCS> ping 192.168.50.10
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=3.591 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=5.208 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=3.803 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=4.044 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=2.442 ms
```

Gambar 59: Sukses Pengujian Koneksi End-to-End Gambar 4

```
VPCS> ping 192.168.50.1
84 bytes from 192.168.50.1 icmp_seq=1 ttl=255 time=7.575 ms
84 bytes from 192.168.50.1 icmp_seq=2 ttl=255 time=2.284 ms
84 bytes from 192.168.50.1 icmp_seq=3 ttl=255 time=2.423 ms
^C
VPCS> ping 192.168.10.11
84 bytes from 192.168.10.11 icmp_seq=1 ttl=62 time=3.340 ms
84 bytes from 192.168.10.11 icmp_seq=2 ttl=62 time=4.027 ms
84 bytes from 192.168.10.11 icmp_seq=3 ttl=62 time=3.950 ms
84 bytes from 192.168.10.11 icmp_seq=4 ttl=62 time=2.928 ms
^C
VPCS> ping 192.168.20.11
84 bytes from 192.168.20.11 icmp_seq=1 ttl=62 time=3.628 ms
84 bytes from 192.168.20.11 icmp_seq=2 ttl=62 time=3.822 ms
84 bytes from 192.168.20.11 icmp_seq=3 ttl=62 time=3.400 ms
84 bytes from 192.168.20.11 icmp_seq=4 ttl=62 time=4.134 ms
^C
VPCS> ping 192.168.30.10
84 bytes from 192.168.30.10 icmp_seq=1 ttl=62 time=4.126 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=62 time=2.618 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=62 time=4.254 ms
84 bytes from 192.168.30.10 icmp_seq=4 ttl=62 time=3.323 ms
84 bytes from 192.168.30.10 icmp_seq=5 ttl=62 time=3.156 ms
^C
```

Gambar 60: Sukses Pengujian Koneksi End-to-End Gambar 5

1.6 Praktikum 5: Pengujian Failover OSPF

Uji ketahanan jaringan (*failover*) dilakukan dengan mematikan interkoneksi utama untuk melihat keandalan protokol OSPF dalam mengalihkan rute trafik ke jalur cadangan secara dinamis.

1. Pengujian Jalur Kondisi Normal (Dua Tautan Cisco-Huawei Aktif)

Memeriksa tabel rute dan rute pelacakan paket (*trace*) menuju LAN Huawei. Paket data seharusnya melewati jalur langsung Cisco → Huawei.

```
CiscoRouter# show ip route 192.168.50.0
```

```
PC-Finance> trace 192.168.50.10
```

```
VPCS> ping 192.168.40.10
84 bytes from 192.168.40.10 icmp_seq=1 ttl=62 time=3.816 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=62 time=4.136 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=62 time=3.909 ms
84 bytes from 192.168.40.10 icmp_seq=4 ttl=62 time=3.406 ms
84 bytes from 192.168.40.10 icmp_seq=5 ttl=62 time=4.513 ms
VPCS>
```

Gambar 61: Hasil Verifikasi Tracing Normal Melewati Jalur Langsung Utama

2. Pengujian Jalur Kondisi Satu Tautan Utama Putus (1 Link Cisco-Huawei Down)

Menonaktifkan salah satu jalur utama (*gi0/3*), kemudian menguji ulang pergerakan paket data.

```

CiscoRouter(config)# interface gi0/3
CiscoRouter(config-if)# shutdown
CiscoRouter(config-if)# exit

CiscoRouter# show ip route 192.168.50.0
PC-Finance> trace 192.168.50.10

```

3. Pengujian Jalur Kondisi Seluruh Tautan Utama Putus (2 Link Cisco-Huawei Down)

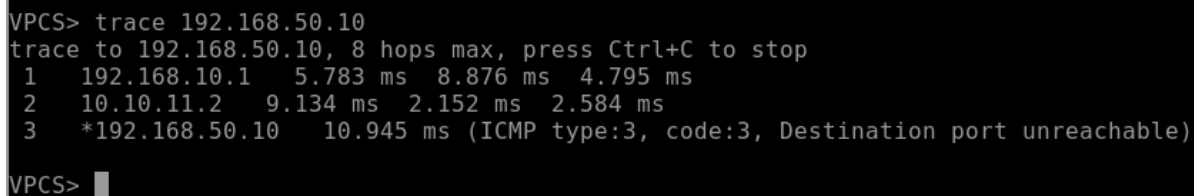
Memutuskan jalur interkoneksi utama kedua (gi0/0). Protokol OSPF harus memindahkan arah rute melalui jalur cadangan MikroTik.

```

CiscoRouter(config)# interface gi0/0
CiscoRouter(config-if)# shutdown
CiscoRouter(config-if)# exit

CiscoRouter# show ip route 192.168.50.0
PC-Finance> trace 192.168.50.10

```



```

VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.10.1    5.783 ms  8.876 ms  4.795 ms
 2  10.10.11.2     9.134 ms  2.152 ms  2.584 ms
 3  *192.168.50.10 10.945 ms (ICMP type:3, code:3, Destination port unreachable)
VPCS>

```

Gambar 62: Pengujian Pemulihan Jalur Kondisi Tautan Utama Aktif Kembali 5

4. Pengujian Pemulihan Jalur Kondisi Tautan Utama Aktif Kembali (Preemption)

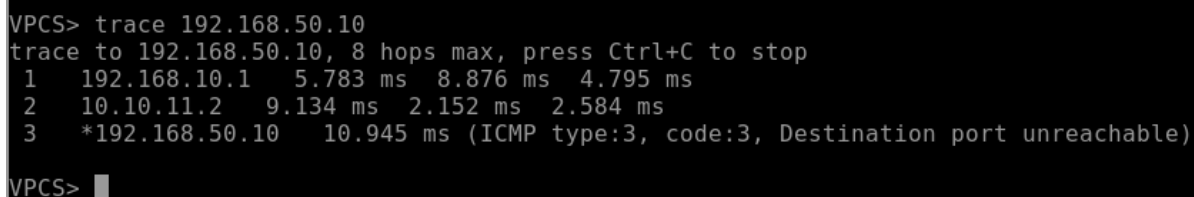
Menghidupkan kembali kedua antarmuka utama. Jalur *routing* otomatis berpindah kembali (*pre-empt*) ke tautan utama karena memiliki nilai *cost* yang lebih rendah.

```

CiscoRouter(config)# interface gi0/3
CiscoRouter(config-if)# no shutdown
CiscoRouter(config-if)# interface gi0/0
CiscoRouter(config-if)# no shutdown
CiscoRouter(config-if)# exit

CiscoRouter# show ip route 192.168.50.0
PC-Finance> trace 192.168.50.10

```



```

VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.10.1    5.783 ms  8.876 ms  4.795 ms
 2  10.10.11.2     9.134 ms  2.152 ms  2.584 ms
 3  *192.168.50.10 10.945 ms (ICMP type:3, code:3, Destination port unreachable)
VPCS>

```

Gambar 63: Pengujian Pemulihan Jalur Kondisi Tautan Utama Aktif Kembali

2 Analisis Hasil Percobaan

Percobaan pada Modul 5 ini berfokus pada konfigurasi dan analisis integrasi jaringan menggunakan VLAN, Trunking, dan protokol *routing* dinamis OSPF pada lingkungan *MultiVendor* (Cisco, Huawei, dan MikroTik). Tujuan utamanya adalah melihat bagaimana segmentasi jaringan lokal dan jalur cadangan (*failover*) dapat berjalan lancar meskipun menggunakan perangkat dengan sistem operasi yang berbeda.

Pada tahap awal, konfigurasi VLAN 10, 20, 30, dan 40 diterapkan pada Cisco Switch untuk memisahkan domain *broadcast* berdasarkan divisi masing-masing, yaitu Finance, HR, IT, dan Marketing. Agar lalu lintas data dari seluruh VLAN ini bisa dikirimkan ke router melalui satu kabel fisik saja, antarmuka penghubung dikonfigurasi sebagai *Trunk Port* dengan enkapsulasi *dot1q*. Proses *inter-VLAN routing* kemudian diatur pada Cisco Router menggunakan metode *Router-on-a-Stick* (ROAS) dengan membuat beberapa *sub-interface*. Selain menjadi *gateway*, Cisco Router juga bertindak sebagai DHCP Server untuk memberikan IP otomatis kepada PC di VLAN 10 and 20. Sementara itu, PC di VLAN 30 dan 40 dikonfigurasi menggunakan IP statis sesuai dengan tabel *addressing*. Pengujian awal menunjukkan seluruh PC di bawah kendali Cisco Switch sudah bisa saling terhubung dan melakukan *ping* ke *gateway* masing-masing.

Tahap berikutnya adalah mengaktifkan *routing* dinamis OSPF Area 0 (*backbone*) untuk menghubungkan Cisco Router, Huawei Router, dan MikroTik. Meskipun ketiga perangkat ini menggunakan vendor yang berbeda, OSPF terbukti memiliki kompatibilitas yang baik sehingga hubungan ketetanggaan (*adjacency*) tetap dapat terbentuk. Status ini diverifikasi melalui terminal masing-masing perangkat dengan perintah `show ip ospf neighbor` di Cisco, `display ospf peer` di Huawei, dan `/routing ospf neighbor print` di MikroTik. Untuk mengatur lalu lintas data, nilai *OSPF cost* pada tiap antarmuka diubah secara manual. Tautan langsung antara Cisco dan Huawei diberi *cost* 10 agar menjadi jalur utama (*primary path*), sedangkan tautan yang mengarah ke MikroTik diberi *cost* 100 sebagai jalur cadangan (*backup path*).

Uji ketahanan jaringan dilakukan menggunakan perintah `trace` untuk melihat rute perjalanan paket data. Dalam kondisi normal, paket data dari PC VLAN menuju LAN Huawei mengalir langsung melalui jalur utama Cisco-Huawei karena nilai metriknya lebih kecil. Ketika salah satu atau kedua jalur utama dinonaktifkan (*shutdown*), protokol OSPF langsung melakukan konvergensi ulang dan mengalihkan lalu lintas data melewati MikroTik sebagai jalur cadangan sehingga konektivitas antar-PC tidak terputus. Saat jalur utama diaktifkan kembali (*no shutdown*), lalu lintas data otomatis berpindah lagi ke jalur langsung Cisco-Huawei karena OSPF mendeteksi rute tersebut kembali efisien. Secara keseluruhan, hasil percobaan menunjukkan bahwa kombinasi VLAN dan OSPF mampu menciptakan infrastruktur jaringan yang stabil, adaptif, dan memiliki redundansi yang baik dalam lingkungan *multi-vendor*.

3 Hasil Tugas Modul

Tugas modul ada di dalam file Readme kelompok

4 Kesimpulan

Berdasarkan hasil percobaan yang telah dilakukan pada Modul 5, dapat disimpulkan beberapa poin berikut:

1. Penerapan VLAN dan jalur *Trunk* berhasil mengisolasi serta mengelompokkan lalu lintas data berdasarkan kebutuhan divisi, sementara metode *Router-on-a-Stick* (ROAS) efektif dalam menghemat penggunaan antarmuka fisik pada router.
2. Protokol OSPF memiliki standar terbuka dengan kompatibilitas yang baik, sehingga mampu menghubungkan perangkat dari berbagai vendor berbeda (Cisco, Huawei, dan MikroTik) untuk saling bertukar informasi rute secara otomatis di area *backbone*.
3. Manipulasi nilai *OSPF Cost* sangat berguna untuk mengatur prioritas jalur trafik. Melalui parameter ini, mekanisme pengalihan jalur cadangan (*failover*) dan pemulihan jalur utama (*preemption*) dapat berjalan otomatis saat terjadi gangguan fisik pada jaringan.